

Ritik Sharma

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Research Interests & Objectives

My research focuses on developing intelligent healthcare systems through AI, biomedical signal processing, and IoT technologies. I am particularly interested in advancing non-invasive diagnostic methodologies using machine learning for real-time health monitoring, with emphasis on creating accessible, cost-effective diagnostic solutions for clinical applications.

Research Areas

- Explainable Artificial Intelligence & Machine Learning
- Deep Learning for Medical Diagnostics
- Biomedical Signal Processing & Analysis
- Human Computer Interaction
- Internet of Things for Healthcare
- Deep Reinforcement Learning for Cybersecurity

Education

Master of Technology (Research)

Jan 2024 – Present

Indian Institute of Technology, Mandi, India

- Cumulative Grade Point Average: 8.75/10.0
- **Relevant Coursework:** Pattern Recognition, Deep Learning and Its Applications, Mobile Virtual Reality and Artificial Intelligence, Introduction to Biosignals, Research Methods and Statistics, Cognitive Psychology
- **Thesis Area:** Development of intelligent breath-based diagnostic systems for diabetes and cardiovascular health monitoring

Bachelor of Technology

Aug 2019 – Aug 2023

Jawaharal Nehru Government Engineering College Sundernagar, Mandi, India

- **Domain:** Electronics and Communication Engineering
- Cumulative Grade Point Average: 8.1/10.0
- **Coursework:** C++, Digital Electronics, Analog Electronics, Signals and Systems, Digital Signal Processing, Analog and Digital Communication, Principles of Soft Computing, Computer Networks

Experience

Project Associate - I, iHub and HCI Foundation, IIT Mandi, India

Jan 2024 – Mar 2025

- Designed and developed a novel AI-enabled Internet of Things (IoT) device for non-invasive diabetes classification, achieving 97.6% diagnostic accuracy through advanced deep learning methodologies
- Developed comprehensive signal processing algorithms and embedded programming solutions, integrating a complete machine learning pipeline using Python, Scikit-learn, and TensorFlow for intelligent healthcare device development
- Extended the diagnostic capabilities to multi-stage diabetes prediction with 90% accuracy and precise blood glucose level estimation (Mean Absolute Error: 17 mg/dL), demonstrating significant clinical utility
- Developed a novel Weighted Average Regressor (WAR) model for blood pressure prediction with less than 8 mmHg error, and Developed a probabilistic ensemble model for HbA1c prediction achieving 83% accuracy

- Architected and deployed a cloud-hosted web application using Streamlit for real-time health monitoring, providing comprehensive diabetes staging, blood glucose level tracking, blood pressure monitoring, and long-term glycemic control assessment

Teaching Assistant, Indian Institute of Technology, Mandi, India

Mar 2025 – Dec 2025

- *Introduction to Biosignals and Cognitive Biomarkers*: Assisted in course delivery, conducted laboratory sessions on signal acquisition and processing, evaluated student assignments, and provided mentorship on biosignal analysis techniques
- *Data Mining for Decision Making*: Supported course instruction, facilitated practical sessions on data mining algorithms and tools, graded assignments and projects, and guided students in applying machine learning techniques for decision-making applications

Publications

Sharma, R., Kapur, R., Kumar, B., Kanwar, V., Bhavsar, A., & Dutt, V. (2025). DiaBreath: A low-cost, non-invasive diabetes monitor via breath. In *ACM Transactions on Computing for Healthcare*. doi:10.1145/3768157

Sharma, R., Sharma, S., Kumar, Y., Kanwar, V., Bhavsar, A., & Dutt, V. (2025). VOCPressNet: An innovative system for breath-based blood pressure prediction. In *Proceedings of the 15th International Conference on Power Electronics and Drive Systems (PEDS)*. doi:10.1109/PEDS63958.2025.11144768

Sharma, S., **Sharma, R.**, Sharma, S., Aarzoo, Kanwar, V., Bhavsar, A., & Dutt, V. (2025). DIAEM-L: Diabetes ensemble analytics model for non-invasive screening in rural populations. In *Proceedings of the 16th International Conference on Computing, Communication and Networking Technologies (ICCCNT)*, Accepted for Publication.

Kapur, R., Kumar, Y., **Sharma, R.**, Sharma, S., Kumar, B., Kanwar, V., Bhavsar, A., & Dutt, V. (2024). GlucoBreath: An IoT, ML, and breath-based non-invasive glucose meter. *IEEE Access*, 12, 65432-65445. doi:10.1109/ACCESS.2024.3392015

Kapur, R., **Sharma, R.**, Kanwar, V., Bhavsar, A., & Dutt, V. (2024, June). SugarControl: an integrative mobile app for type 2 diabetes management through yoga meditation and lifestyle interventions. In *Proceedings of the 17th International Conference on Pervasive Technologies Related to Assistive Environments* (pp. 647–653).

Projects

Advanced Deep Learning Framework for Low-Dose CT Image Denoising

Feb 2025 – Jul 2025

- Developed encoder-decoder architectures with multi-head channel attention for low-dose CT image enhancement, achieving 43 dB PSNR
- Developed novel attention mechanisms to capture local and global features for superior medical image reconstruction
- Deployed Flask-based visualization platform for quantitative model performance assessment

Deep Reinforcement Learning for Cybersecurity: Network Intrusion Detection

Feb 2024 – Jun 2024

- Developed deep reinforcement learning framework for network intrusion detection, achieving 98% accuracy across DoS, Probe, R2L, U2R attacks
- Developed real-time threat detection system using advanced RL algorithms for cybersecurity applications
- Created interactive Streamlit dashboard for network security visualization and monitoring

Comprehensive Non-Invasive Health Monitoring System via Breath Analysis

Jan 2024 – Present

- Developed novel feature engineering methodology for breath-based health diagnostics, achieving 97.6% diabetes classification accuracy
- Engineered ensemble ML pipelines for multi-stage diabetes prediction (90% accuracy), blood glucose estimation (17 mg/dL MAE), and blood pressure prediction (<8 mmHg error)
- Developed HbA1c prediction (83% accuracy) and deployed cloud-based health monitoring platform with real-time visualization

Technical Proficiencies

Frameworks: TensorFlow, PyTorch, Keras, Scikit-learn, Streamlit

Research Tools & Methodologies: Statistical Analysis, Signal Processing, Feature Engineering, ML Model Development, IoT Systems Development, Git, LaTeX | **Languages:** Python, C++